Notes: Merton (JF, 1987)

A Simple Model of Capital Market Equilibrium with Incomplete Information

Contents

1 Big picture

- **Question.** How does incomplete diffusion of information across investors change equilibrium asset prices, expected returns, and firms' investment choices, even when investors are rational and markets are otherwise frictionless?

- **Setting.** Merton builds a two-period capital market equilibrium model in which each investor knows only a subset of the available securities. Investors are otherwise standard price-taking mean-variance optimizers.

- **Core challenge.** In the standard Sharpe-Lintner-Mossin CAPM, all investors know all securities, so the market portfolio is mean-variance efficient and expected returns depend only on market beta. But in reality, investors hold only a tiny subset of available assets. Merton asks whether this limited "investor recognition" alone can generate systematic pricing effects.

- **Main methodological idea.** The model inserts incomplete information in a very specific way: investor j uses security k in the portfolio only if the investor knows that security's parameters. This generates a shadow cost of not being known, which behaves like an additional required return.

- **Main theoretical result.** Incomplete information adds a security-specific premium to expected returns. With market beta β_k and security-specific shadow cost λ_k, the pricing relation becomes

  $$\bar{R}_k - R = \beta_k(\bar{R}_M - R) + \lambda_k - \beta_k \lambda_M,$$

  so the CAPM alpha is

  $$\alpha_k = \lambda_k - \beta_k \lambda_M.$$

  Securities with smaller investor bases have larger λ_k and therefore higher expected returns and lower prices.

- **Firm-value implication.** If all investors knew the firm, its value would be V_k^*. Under incomplete information,

  $$V_k = \frac{V_k^*}{1 + \lambda_k / R},$$

  so incomplete information acts like an additional discount rate.

- **Main empirical intuition.** The model rationalizes several anomalies often viewed as problems for the CAPM: the Security Market Line can be "too flat," alpha can depend on total variance, and neglected or less well-known firms can earn higher returns.

- **Economic takeaway.** Investor recognition is an asset-pricing state variable. A firm that is followed by fewer investors faces a higher cost of capital, and managers therefore have an incentive to expand the firm's investor base through visibility, publicity, ratings, and exchange listing.

2 Model environment and key objects

2.1 This is mainly a theory paper: what are the key objects?

- There is no modern structural estimation or reduced-form dataset at the core of the paper. The "data" objects are mostly model primitives and one small cross-sectional shareholder table used for quantitative intuition.
- The central objects are:
  - firm cash-flow parameters (I_k, μ_k, a_k, s_k),
  - a common factor $\tilde{Y}$ and idiosyncratic shocks $\tilde{\varepsilon}_k$,
  - investor information sets J_j,
  - the number of investors who know security k, denoted N_k,
  - the investor-recognition share
$$q_k = \frac{N_k}{N},$$
    and
  - the shadow cost of incomplete information λ_k.
- The whole paper is organized around the idea that the cross-sectional variation in q_k changes equilibrium returns and values.

2.2 Firms, cash flows, and return structure

- There are n firms. End-of-period cash flow of firm k is
$$\tilde{C}_k = I_k[\mu_k + a_k\tilde{Y} + s_k\tilde{\varepsilon}_k].$$
- $\tilde{Y}$ is a common factor with mean zero and variance one. The firm-specific shock $\tilde{\varepsilon}_k$ is orthogonal to $\tilde{Y}$ and to all other firms' idiosyncratic shocks.
- Let V_k be firm value at the beginning of the period and let $\tilde{R}_k = \tilde{C}_k/V_k$ be the return per dollar invested in the firm. Then returns take the one-factor form
$$\tilde{R}_k = \bar{R}_k + b_k\tilde{Y} + \sigma_k\tilde{\varepsilon}_k,$$
  where
$$\bar{R}_k = \frac{I_k\mu_k}{V_k}, \qquad b_k = \frac{a_k I_k}{V_k}, \qquad \sigma_k = \frac{s_k I_k}{V_k}.$$
- This is the same return structure as a one-factor diagonal model or one-factor APT environment. The novelty is not the factor structure itself; it is the information structure across investors.

2.3 Additional traded securities

- Besides the n firms, there are two additional securities:

- a riskless security with sure gross return R,
- a security combining the riskless asset with a forward contract on the common factor, whose return is normalized to

$$\tilde{R}_{n+1} = \bar{R}_{n+1} + \tilde{Y}.$$

- These two securities are assumed to be inside securities, so their aggregate demand must be zero in equilibrium.

2.4 Investors and information sets

- There are N investors, sufficiently many and sufficiently small that each is a price taker.
- Investor j has mean-variance preferences over end-of-period wealth:

$$U_j = \mathbb{E}(\tilde{R}^j W^j) - \frac{\delta_j}{2W^j}\mathrm{Var}(\tilde{R}^j W^j).$$

- An investor is said to be *informed about security* k if he knows $(\bar{R}_k, b_k, \sigma_k^2)$.
- Let J_j denote the set of securities known by investor j. All investors know the riskless asset and the factor-mimicking security, but they generally know only a subset of the n firms.
- **Key behavioral assumption:** investor j uses security k in the optimal portfolio only if $k \in J_j$.
- Interpretation: there is a fixed "set-up" cost to becoming aware of and processing information about a security. This is Merton's formalization of investor recognition.

2.5 Investor recognition and the shadow cost of incomplete information

- Define

$$q_k = \frac{N_k}{N}$$

  as the fraction of investors who know about security k.
- The smaller is q_k, the narrower the investor base and the more "neglected" the security.
- In equilibrium, this limited recognition shows up as a shadow cost λ_k of incomplete information. This shadow cost is zero only when everyone knows the security.
- The whole paper can be read as turning a vague idea—"neglected stocks require higher returns"—into a formal equilibrium object.

3 Models and empirical specifications (all equations + interpretation)

3.1 Investor problem and portfolio demand

Define the excess-return object

$$\Delta_k \equiv \bar{R}_k - R - b_k(\bar{R}_{n+1} - R).$$

This is the part of expected return on security k that is *not* explained by its loading on the common factor.

- Investor j solves a constrained mean-variance problem where the Kuhn-Tucker multiplier λ_k^j enforces the "do not hold what you do not know" restriction.
- The first-order conditions imply optimal common-factor exposure and risky-asset holdings:

$$b^j = \frac{\bar{R}_{n+1} - R}{\delta_j},$$

$$w_k^j = \frac{\Delta_k}{\delta_j \sigma_k^2}, \qquad k \in J_j,$$

and

$$w_k^j = 0, \qquad k \in J_j^c.$$

- Interpretation:
  - All investors choose the same factor exposure formula, given risk aversion.
  - Conditional on knowing a security, the desired holding is increasing in its factor-adjusted excess return Δ_k and decreasing in idiosyncratic variance σ_k^2.
  - Unknown securities are literally excluded from the investor's opportunity set.
- The shadow cost for not knowing security k is

$$\lambda_k^j = \Delta_k, \qquad k \in J_j^c.$$

Merton stresses that this shadow cost is measured in units of expected return: it is the premium an investor forgoes because the security is outside the recognized set.

3.2 Market clearing and equilibrium expected returns

- Aggregate demand for security k is

$$D_k = N_k W \frac{\Delta_k}{\delta \sigma_k^2},$$

under the simplifying assumption that investors have identical preferences and equal wealth.

- Since equilibrium requires $V_k = D_k$, the market-share weight of security k satisfies

$$x_k \equiv \frac{V_k}{M} = \frac{q_k \Delta_k}{\delta \sigma_k^2},$$

where M is aggregate wealth.

- Rearranging gives the expected return on security k:

$$\bar{R}_k = R + b_k b \delta + \frac{\delta x_k \sigma_k^2}{q_k}.$$

- Interpretation: expected return has two pieces.
  - The first piece, $b_k b\delta$, is the standard factor-risk compensation.
  - The second piece, $\delta x_k \sigma_k^2/q_k$, is the incomplete-information premium. It is larger when the firm has high firm-specific risk, large market weight, and a small investor base.

3.3 Firm value under incomplete information

- Substituting the return formulas into the firm's valuation gives

$$V_k = \frac{I_k\left[\mu_k - \delta b a_k - (\delta s_k^2 I_k)/(q_k M)\right]}{R}.$$

- Holding aggregate variables fixed, if all investors were informed about firm k, value would be V_k^*. With incomplete information,

$$V_k = V_k^* - \frac{\delta(1-q_k)s_k^2 I_k^2}{q_k M R}.$$

- Define the aggregate shadow cost per investor for security k as

$$\lambda_k = (1-q_k)\Delta_k.$$

- Then the price impact can be written compactly as

$$V_k = \frac{V_k^*}{1+\lambda_k/R}.$$

- Interpretation: incomplete information lowers price exactly as if the firm were being discounted at an additional rate λ_k.
- Moreover,

$$\bar{R}_k - \bar{R}_k^* = \lambda_k\left(\frac{\bar{R}_k^*}{R}\right),$$

so incomplete information raises required return proportionally to the shadow cost.

3.4 Modified Security Market Line

- Define market beta in the usual way:

$$\beta_k = \frac{b_k b + x_k \sigma_k^2}{\mathrm{Var}(\tilde{R}_M)}.$$

- Then equilibrium expected returns satisfy

$$\bar{R}_k = R + \delta\mathrm{Var}(\tilde{R}_M)\beta_k + \lambda_k.$$

- Averaging across all securities implies

$$\bar{R}_M - R = \delta \mathrm{Var}(\tilde{R}_M) + \lambda_M,$$

where

$$\lambda_M \equiv \sum_{k=1}^{n} x_k \lambda_k$$

is the market-weighted average shadow cost.

- Substituting gives the key pricing equation:

$$\bar{R}_k - R = \beta_k(\bar{R}_M - R) + \lambda_k - \beta_k \lambda_M.$$

- Therefore the CAPM alpha is

$$\alpha_k = \lambda_k - \beta_k \lambda_M.$$

- Interpretation:
  - Under complete information, $\lambda_k = 0$ for all k, so alpha is zero and the market portfolio is mean-variance efficient.
  - Under incomplete information, the market portfolio is *not* mean-variance efficient with respect to all available securities.
  - A fully informed investor wants to tilt away from the market portfolio by an amount proportional to $\lambda_k/(\delta\sigma_k^2)$.

3.5 Comparative statics: what raises expected returns?

- Merton computes the elasticity of expected excess return with respect to four primitives: common-factor exposure b_k, firm size x_k, firm-specific variance σ_k^2, and investor recognition q_k.
- The signs are simple:

$$\psi(b_k) > 0, \qquad \psi(x_k) > 0, \qquad \psi(\sigma_k^2) > 0, \qquad \psi(q_k) < 0.$$

- Interpretation:
  - higher common-factor exposure $\Rightarrow$ higher expected return,
  - larger firm-specific variance $\Rightarrow$ higher expected return,
  - larger firm size $\Rightarrow$ higher expected return *conditional on the other objects*,
  - larger investor base (better recognition) $\Rightarrow$ lower expected return.
- The last result is the central one: better-known firms have lower required returns because more investors stand ready to hold them.

3.6 Cross-sectional predictions for anomalies

- If one sorts securities only by beta, Merton's model predicts a "too flat" Security Market Line. In a sample where securities share the same q_k, σ_k^2, and x_k but differ in beta,

$$\frac{\partial \alpha_k}{\partial \beta_k} = -\lambda_M < 0.$$

  So alpha declines with beta.

- Aggregating over such a sample yields

$$\mathbb{E}(\alpha_k \mid \beta_k) = (1 - \beta_k)\lambda_M,$$

  which resembles the logic of the Black zero-beta CAPM, but for a different reason.

- Merton then shows that alpha should depend on more than beta. Holding market risk fixed and assuming $q_k < 1$,
  - firms with higher firm-specific variance should have higher alpha,
  - firms with larger size should have higher alpha *conditional on equal volatility and equal recognition*,
  - firms with larger investor bases should have lower alpha.

- This helps organize several anomalies:
  - alpha depending on total variance,
  - neglected-firm effects,
  - the idea that small-firm effects may weaken once one controls for investor recognition or institutional following.

3.7 Firm behavior and endogenous investor base

- Because a larger investor base reduces expected return and increases firm value, managers have an incentive to enlarge the firm's investor base.

- Merton explicitly interprets this as spending on investor visibility: publicity, broad media coverage, listing on major exchanges, obtaining outside ratings, and investor-targeted advertising.

- Let $F(N_k)$ denote the cost of creating an investor base of size N_k, with $F'(N_k) > 0$ and $F''(N_k) > 0$.

- The optimal investor-base choice solves

$$F'(N_k) = \frac{\delta s_k^2 I_k^2}{R N_k^2 W}.$$

- Interpretation: firms with greater firm-specific risk have more to gain from being widely recognized, because recognition lowers the information premium.

- Merton also studies the firm's investment decision. Let $\psi(I_k)$ denote the elasticity of expected excess return with respect to firm investment. He shows that for firms with relatively small investor bases and high firm-specific risk, the required return rises materially when the firm expands, so these firms face significantly downward-sloping investor demand curves.
- The joint choice of investment and investor base implies

$$I_k = \frac{N_k W[\mu_k - R - \delta b a_k]}{2\delta s_k^2},$$

and, substituting back,

$$F'(N_k) = \frac{W[(\mu_k - R - \delta b a_k)/s_k]^2}{4\delta R}.$$

- Interpretation: production and investor recognition are jointly determined. More investment tends to go together with a broader investor base, and a broader investor base lowers the cost of capital and supports more investment.

4 Identification and instruments (what makes the results credible)

4.1 What generates departures from the CAPM?

- The paper's departures from the CAPM come from a single structural friction: different investors know different securities.
- There is no irrationality, no noise trading, and no strategic exploitation of uninformed investors within a security. The pricing wedge is generated by limited recognition, summarized by q_k and λ_k.
- This makes the source of the anomaly clean: if a return pattern cannot be traced back to market beta alone, the model asks whether it can be traced back to investor recognition.

4.2 Why the comparative statics are informative

- The model produces sign predictions that are testable in cross section:
  - lower q_k should be associated with higher expected returns and lower values,
  - alpha should depend on firm-specific variance and investor recognition,
  - the relation between alpha and size is conditional, not unconditional.
- This matters because Merton is not just saying "information might matter." He is saying exactly how incomplete information should distort the Security Market Line and firm valuation.

4.3 Why Table 1 matters

- The paper's one table is not a full empirical test, but it is important. It shows that investor recognition plausibly varies a lot across firms and that this variation is not the same thing as raw market size.

Table I
Distributions of Market Value and Market Value Per Shareholder
(1387 firms)

	December 31, 1985			
Group	Market Value of Group/ Market Value of Sample (%)	Cumulative Market Value (%)	Market Value per Shareholder/ Average for Sample	Indicated q_k/q_{10}
1	0.11	0.11	0.161	.02
2	0.33	0.44	0.371	.03
3	0.69	1.13	0.545	.04
4	1.21	2.34	0.793	.05
5	1.97	4.31	0.972	.06
6	3.24	7.55	1.127	.09
7	5.28	12.83	1.257	.13
8	8.65	21.48	1.490	.18
9	15.05	36.53	1.348	.34
10	63.47	100.00	1.957	1.00

- That is the key empirical precondition for the model to matter: if every firm were equally recognized, then the whole q_k channel would disappear.

4.4 Limits of the model

- The return process is one-factor and especially simple.
- Informed investors who know a given security all share the same beliefs about that security.
- The paper abstracts from asymmetric-information trading games, transactions costs, taxes, and institutional detail except insofar as they can be represented as affecting whether investors know a stock.
- Merton is explicit that the model is chosen for transparency, not realism in every detail. The point is to show that modest information frictions can already explain behavior often labeled anomalous.

4.5 Relation to the literature

- If all investors know all securities ($q_k = 1$ for all k), the model collapses to the standard Sharpe-Lintner-Mossin CAPM.
- Relative to Grossman-Stiglitz, the paper does not model strategic interaction between informed and uninformed traders inside a security. Instead, it studies incomplete diffusion of recognition across securities.
- Relative to Bawa-Klein-Barry-Brown differential-information models, Merton's emphasis is not on different precision of signals across securities but on different breadth of investor cognizance.
- Relative to the empirical neglected-stock literature, Merton supplies a formal equilibrium mechanism linking recognition to higher required return and lower firm value.

5 Tables

5.1 Table 1: distributions of market value and market value per shareholder

Read:

- The paper ranks 1,387 firms into 10 groups by market value and compares two objects: each decile's market share and its market value per shareholder.
- The top decile accounts for **63.47%** of total market value, and the top two deciles together account for nearly **80%**. So market value is extremely concentrated.
- But market value per shareholder is much less concentrated. The top decile is only **1.957** times the sample average, while the bottom decile is **0.161** times the sample average. So the top-to-bottom ratio is only about 12, much smaller than the ratio for total market value.
- Merton interprets the last column, the indicated q_k/q_{10}, as rough evidence that the lower 80% of firms have substantially smaller investor-recognition shares than the largest firms.
- The table therefore supports the key premise of the model: differences in investor recognition are large enough to matter in practice, and they are not mechanically the same thing as size.

5.2 No formal figures: the key "visuals" are equations (20), (26), and (37)

Read:

- **Equation (20)** is the valuation result:

$$V_k = \frac{V_k^*}{1 + \lambda_k/R}.$$

  This is the cleanest statement that incomplete information acts like an extra discount rate.

- **Equation (26)** is the asset-pricing result:

$$\alpha_k = \lambda_k - \beta_k \lambda_M.$$

  This is the equation to remember if you want to know how the model breaks the CAPM.

- **Equation (37)** is the firm-behavior result:

$$F'(N_k) = \frac{W[(\mu_k - R - \delta b a_k)/s_k]^2}{4\delta R}.$$

  This equation endogenizes investor recognition by showing when a firm will pay to broaden its investor base.

6 Short summary

- **Method.** Merton modifies a one-factor CAPM environment by assuming that each investor knows only a subset of securities and invests only in the securities he knows.
- **Pricing result.** Incomplete information creates a shadow cost of limited investor recognition. This raises expected returns, lowers prices, and generates nonzero alpha:

$$\alpha_k = \lambda_k - \beta_k \lambda_M.$$

- **Cross-sectional implication.** Better-known firms have lower expected returns. Alpha can depend on investor recognition and firm-specific variance, so the Security Market Line can appear too flat.
- **Firm implication.** Managers optimally spend resources to enlarge the investor base because broader recognition lowers the cost of capital and raises firm value.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- A rational asset-pricing model with only modest informational incompleteness can generate systematic departures from the complete-information CAPM.
- These departures are not random. They take a precise form tied to investor recognition, firm-specific variance, and market beta.
- The paper also shows that asset pricing and corporate finance are linked: investor recognition affects not just returns but also firm value and real investment decisions.

7.2 Economic intuition

- A stock followed by fewer investors must be absorbed by a narrower clientele. That makes it effectively harder to place in portfolios and raises the required return investors demand.
- Because the required return is higher, the stock price is lower. Since the firm's cost of capital is higher, the firm invests less unless it can broaden the investor base.
- In this way, investor recognition plays a role similar to a demand-side friction in capital markets: the fewer the investors who recognize the stock, the steeper the effective demand curve facing the firm.

7.3 Takeaway

This paper shows how far one can get without abandoning rational asset pricing. The main message is not that CAPM anomalies require irrationality, but that they may reflect omitted institutional and informational frictions. In Merton's model, being well known lowers a firm's cost of capital. Investor recognition therefore becomes a central bridge between asset pricing anomalies, neglected-stock effects, and firm behavior.